GraphVOI: allocation-aware clarification for sequential reciprocal matching

Round 1 research note, The Sequential Matching Problem (Romeo & Juliet / Vouchsafe)

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Research question. When clarification has a cost, should we ask the most uncertain question, or the question whose answer is most likely to improve the best feasible allocation?

1 Summary

This problem is not only about predicting whether two people are compatible. The policy works on a pool that keeps changing, it has to respect reciprocal hard constraints, it chooses clarification requests under a daily budget, it introduces non-overlapping pairs, and it learns from outcomes that often show up weeks later. The spec itself says that ranking pairs one by one is not enough, because choosing one edge changes the options of everyone else [1].

We think the same applies to questions. Asking about someone is only useful if the answer could change who gets introduced. Our proposal is built around this one hypothesis, and we call the method **GraphVOI**. A lightweight pair model gives edge utilities, the supplied eligibility rules decide which edges are valid or still unresolved, and a maximum-weight matching solver picks the final batch. Waiting is handled at first with a simple conservative threshold, not a separate learned RL policy.

Our background is mostly reinforcement learning, so the first thing we considered was an end-to-end sequential policy. We dropped it. The action space is a matching, rewards are rare and delayed, feedback only exists for pairs we choose, and each call gets 10 seconds on 2 CPUs. The matching part has a clean structure that a solver handles exactly, so we keep learning for the parts that are actually uncertain: what to ask, and later possibly when to wait.

We do not claim that value of information, graph matching, bandits or RL are new on their own. What we want to test is the link between information gathering and the pool-level matching objective.

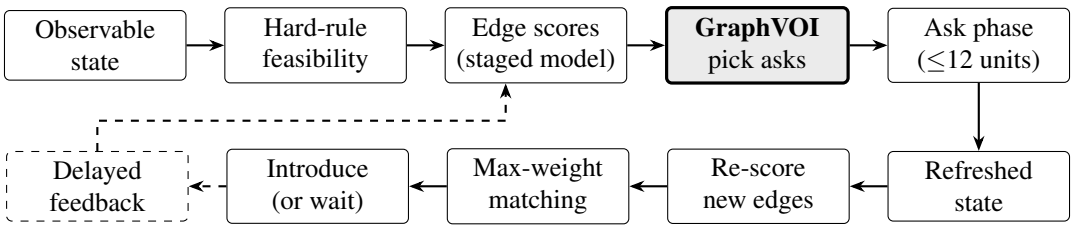


Figure 1: One day of our policy. Only the shaded box is the research part; the rest is standard and can be replaced by something simpler.

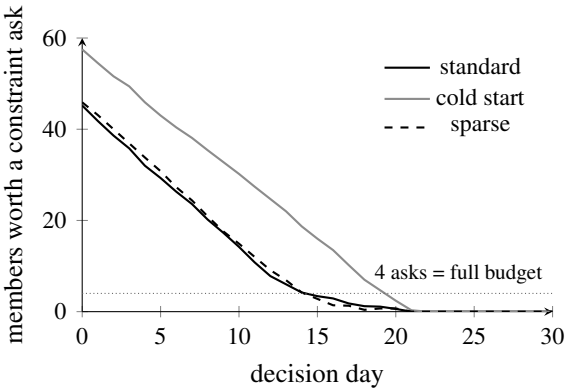
2 Problem interpretation

The evaluator asks for two decisions every day: what to clarify, and which valid pairs to introduce. Missing hard information blocks an edge, and a chosen pair uses up both people for that batch. Feedback is selective, since only introduced pairs produce evidence, and recent outcomes are right-censored. The official score is MSMI per 100 arrived members, averaged within and then across six scenario families [1].

Decision	How we handle it
Reciprocal feasibility	Supplied hard-rule semantics. A learned score never overrides a hard failure, and missing hard data blocks the pair until it is resolved.
Pair utility	Small calibrated model on observable pair and market features. Probability output is optional; if we report it, "A accepts B" and "both accept" stay separate.
Clarification	GraphVOI: estimate whether resolving a missing field changes the best allocation enough to be worth it. Asks are a constraints bundle (3 units) or one soft field (1 unit), 12 units a day.
Allocation	Maximum-weight general-graph matching over currently feasible edges.
Waiting	A simple day-dependent threshold first. RL or bandit timing is only a later experiment, if the core works.
Delayed data	Train only on outcome windows that have matured; unresolved cases stay censored instead of being labelled as failures.

3 What we found when we ran the starter kit

Before fixing the method, we ran the official greedy baseline on the public simulator (10 seeds for each of the six variants), mainly to check that clarification really has room to matter. A few results changed our plan.



Variant	Edges	Used	Mutual	MSMI
standard	119	92	13.0	1.2
sparse	22	22	3.9	0.2
cold start	99	80	9.8	0.6
delayed	119	90	13.4	0.8
shift	119	93	12.2	1.1
drift	119	92	12.5	1.1

Per episode, mean of 10 seeds. Edges: feasible pairs by day 59. Used: introduced by greedy.

Figure 2: Greedy baseline on the public simulator. Left: available members who still have missing (not declined) hard fields. Right: graph size and outcomes per episode.

- **The constraint budget is only tight at the start.** After about three weeks nobody is left who could usefully get a constraint ask. If GraphVOI only ranked constraint bundles, it would matter for roughly the first third of the episode. So we let it rank soft-field questions too.
- **About a third of members are blocked for good.** 32% of members (39% in cold start) declined at least one hard field. Every pair involving them stays blocked, and asking them again cannot help.
- **The graph is small.** Around 120 feasible edges by day 59 in most variants, only 22 in sparse geography, and greedy uses most of them. An exact max-weight matching takes about 13 ms, so re-solving it many times per call is affordable.
- **The target outcome is rare.** Greedy gets about one MSMI pair per episode, and 25 of our 60 runs had none. Comparing methods on MSMI alone will be noisy, which shapes the experiment plan in Section 7.

4 Main hypothesis

Uncertainty-first clarification can waste budget. A field can be very uncertain and still have nothing to do with today's useful matchings. On the other hand, a single unresolved hard field can matter a lot when it sits on an edge next to a scarce member, or when it changes which pairs can exist together.

H1. With equal ask budgets, GraphVOI clarification gives a higher policy-level MSMI than uncertainty-first clarification.

H1b (secondary, fixed in advance). The same comparison on mutual acceptances per 100 arrived members. It has about ten times more events and is the official second tie-breaker.

This is the central claim. If it does not hold consistently across seeds and scenario families, the hypothesis is not supported, and we will report that result instead of hiding it behind a more complicated model.

5 Research grounding

The pieces we use already exist; what we put together is the link to the competition's clarification action.

Matching when information costs something. Blum et al. [3] study stochastic matching where edges must be queried before use, motivated by kidney exchange. Singla [4] subtracts the cost of probing from the value of a maximum-weight matching. Bowers and Waggoner [5] study two-sided matching where each side of an edge has a value that costs something to inspect. Chawla et al. [6] give a general framework for selection problems with costly information. These works show that a query should be judged by what it does to the final matching, which is the principle we build on.

Asking questions in matching markets. Drummond and Boutilier [7] choose preference queries for stable matching using regret, so they only ask what matters for the matching. NBQ [8] picks matchmaking interview questions by expected information gain about a person's profile. That is close to the uncertainty-first rule we compare against. Reciprocal recommender work explains why both directions of a pair must be modelled [9].

Sequential learning. Contextual matching-bandit work shows that small context changes can completely change the optimal matching [10], which is why the value of one local fact depends on the rest of the pool. Delayed-feedback bandits treat outcomes that have not arrived yet as delayed or censored, not as negatives [11].

These papers motivate our structure, but they do not show that GraphVOI will help in this simulator. That is an empirical question, and we treat the literature as a reason for the components, not as proof of leaderboard gains.

6 Proposed method

6.1 Feasibility and edge utility

For every currently relevant pair (i, j) we first apply the official reciprocal eligibility logic and drop pairs already introduced. Known violations remove the edge. Missing hard information leaves an unresolved edge that cannot be introduced yet. For feasible pairs, a lightweight model estimates an edge utility from observable soft fields, waiting time, pair history and simple market features such as feasible degree and best alternative edge.

The utility is built from three short stages: both accept, a date happens, and both give a timely Yes. The first model for each stage will be regularized logistic regression or histogram gradient boosting, trained offline on simulator rollouts with declared seeds. Inside an episode only the first stage is updated, because MSMI labels mature too late to be useful. We want the experiment to test the clarification policy, not to bury it inside model capacity.

6.2 Maximum-weight allocation

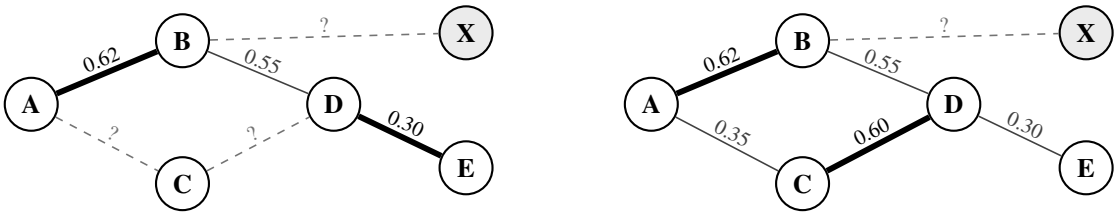
After clarification, the feasible edge utilities define a graph. We solve a maximum-weight matching so that pairs are chosen together instead of greedily. With at most 200 members there are at most 19,900 pairs before pruning, and in practice the feasible graph is far smaller (Section 3), so a standard matching implementation fits easily inside the runtime limits [1, 2]. Before a batch is returned we check it again: nobody appears twice, no pair is repeated, nobody unavailable or with an outstanding introduction is included, and every pair passes the hard rules. One invalid episode makes the whole submission ineligible, so this check is not optional [1].

6.3 GraphVOI

Let $V(G_t)$ be the value of the best current matching under the policy's edge utilities. For a possible clarification q with answer y, we estimate

$$\text{GraphVOI}(q) \;=\; \mathbb{E}_y\big[V(G_t \mid q = y)\big] \;-\; V(G_t) \;-\; \lambda\, c(q).$$

In words: imagine the answer, update the graph, re-solve the matching, and see how much better it gets on average.



Now: best matching $\{AB, DE\}$, value 0.92 — If C turns out compatible: $\{AB, CD\}$, value 1.22

Figure 3: A toy example (values made up). X has all 11 hard fields missing, so an uncertainty-first rule asks about X. But even if X fits B, B is already in a better pair, so nothing changes. C has only two missing fields, yet its answer can change the whole matching. Thick lines are matched pairs, dashed lines are blocked by missing data.

Answer distribution. It is estimated only from lawful training data or simulator rollouts. For a constraints bundle we fit a simple model on fully observed members in the training pools, using age, gender and zone (always visible), and sample a few possible answers. For a soft field, the answer changes edge utilities rather than feasibility. No hidden simulator value is read at inference.

Choosing the day's asks. Unused budget is lost at the end of the day, and clarification cost only enters the official ranking as the third tie-breaker, so we expect λ to be close to zero. In practice we pick asks greedily by GraphVOI per unit until the 12 units are used. If two blocked members are each other's only possible partner, asking just one of them does nothing, so we score that pair of asks as one 6-unit action.

Keeping it fast. We only evaluate GraphVOI for a frontier of unresolved members that touch high-utility edges or bottleneck components. One answer only changes one connected component, so we re-solve just that part. We cap the frontier, cache pair scores, and fall back to a simpler heuristic if a call gets close to the 10-second limit.

6.4 Missing and delayed outcomes

We keep observed, not-asked and declined states separate, and a declined field is never filled in. A recent introduction whose response or later event is still inside its observation window is not used as a negative label. A pair that never had a date has no second-meeting label. Model updates use only matured examples, in line with the spec's warning about right-censoring and selective feedback [1]. The policy process restarts on every call, so the small per-episode statistics we need (member response counts, pending introductions) are carried in the memory field, which has a 1 MiB limit [2].

6.5 Waiting without overcomplicating Round 1

The system may return an empty or partial matching, so waiting matters, especially since each introduction keeps both people busy for at least eight days. In the first build an edge is introduced only if its utility clears a threshold that becomes less strict near the end of the decision horizon. We can also hold back a merely good edge if one endpoint has very few alternatives left (a simple scarcity check on feasible degree and best alternative edge). Both are tuned on training and validation pools. A contextual bandit can be tested later as an ablation, but it is not part of the novelty claim and the core system works without it.

7 Experiments that can falsify the idea

Every variant uses the same pair model, the same matching solver, the same budget and the same seeds; only the part named in the row changes.

Variant	Purpose
Official greedy baseline	Reference supplied by the organisers.
Official no-clarification baseline	Shows how much asking matters at all.
Official random-feasible baseline	Lower reference for allocation quality.
Learned edge model + greedy	Separates prediction gains from allocation gains.
Learned edge model + max-weight matching	Global allocation without our clarification idea.
+ uncertainty-first clarification	Strong generic active-information baseline.
+ degree-first clarification	Uses the graph but not the matching: unlock whoever adds the most feasible edges.
+ GraphVOI clarification	Tests the main hypothesis. Must beat both rows above.
+ GraphVOI + simple waiting threshold	Final intended core system.
Optional: contextual bandit timing	Only if the core is stable. Tests whether learned timing beats the threshold.

The degree-first row matters to us. If GraphVOI beats uncertainty-first but not degree-first, then the matching part of the idea is not what helps, and we will say so.

The primary metric is the official MSMI per 100 arrived members. Because it is so rare, we run at least 100 paired episodes per public variant (the same world for every method) and report bootstrap intervals for MSMI and for mutual acceptances. We also report the full funnel the spec asks for: assignments, mutual acceptances, dates, coverage, first-introduction waiting time with the unserved count, clarification cost, missing feedback, invalid-episode count and runtime. Hyperparameters are chosen on the supplied training and validation pools, and the development-test pools are kept separate.

8 Failure cases and what would change our mind

- GraphVOI may be too short-sighted when a question is only useful several days later. We will test a small future-value bonus, but not add a full RL system unless this failure actually shows up.
- Estimated answer distributions may be poor for rare subgroups. We will shrink toward broader frequencies and avoid high-variance questions.
- When several members compete for the same popular candidate, maximum-weight matching can over-serve edges that look good today. The waiting threshold and scarcity check are meant to reduce this, and we report waiting and coverage next to MSMI.
- In the delayed-dates variant, outcomes arrive even later, so in-episode learning has almost nothing to use. The staged models trained offline carry most of the weight there, and we report this variant separately.
- In sparse geography nearly every feasible edge gets used anyway, so better asking may not matter there. We report each variant separately instead of letting the average hide it.
- Members who declined a hard field can never be introduced. We report how many there are and do not count them as policy failures.
- If GraphVOI does not beat uncertainty-first (and degree-first) clarification across controlled runs, we will not claim that allocation-aware asking helped.
- If the learned pair model adds little, we can keep a rule-based edge score. The GraphVOI hypothesis does not need a deep predictor.

9 Novelty claim, stated narrowly

What we claim. We propose to rank costly clarification actions by their estimated counterfactual effect on the best feasible pool-level matching, rather than by local uncertainty alone, and to test whether that improves the simulator's sequential MSMI objective. As far as we found, the closest work either queries single edges or inspects per-edge values in a static problem [3, 4, 5], or chooses questions by information gain about a profile [8]. What is specific here is that one ask reveals a whole member's constraints, so it can change every edge around that member at once, asks have different costs, the budget resets daily, and outcomes arrive late and censored inside a pool that changes over time. We could not find this combination tested before, but we treat that as our reading of the literature, not a guarantee.

What we do not claim. That value of information, maximum-weight matching, contextual bandits, reinforcement learning or GNNs are new; that the synthetic simulator represents real relationship outcomes; or that the method is guaranteed to improve the score before controlled evaluation.

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